Post-processing of Abaqus results with the A_ scripts

Section export, image export and pushover curves — one module of the seismic analysis workflow

Scripts: `Abaqus-scripts-for-output-processing / 3_export_and_image_export` · Abaqus/CAE or Viewer 2021 (Python 2.7) · 10 November 2026

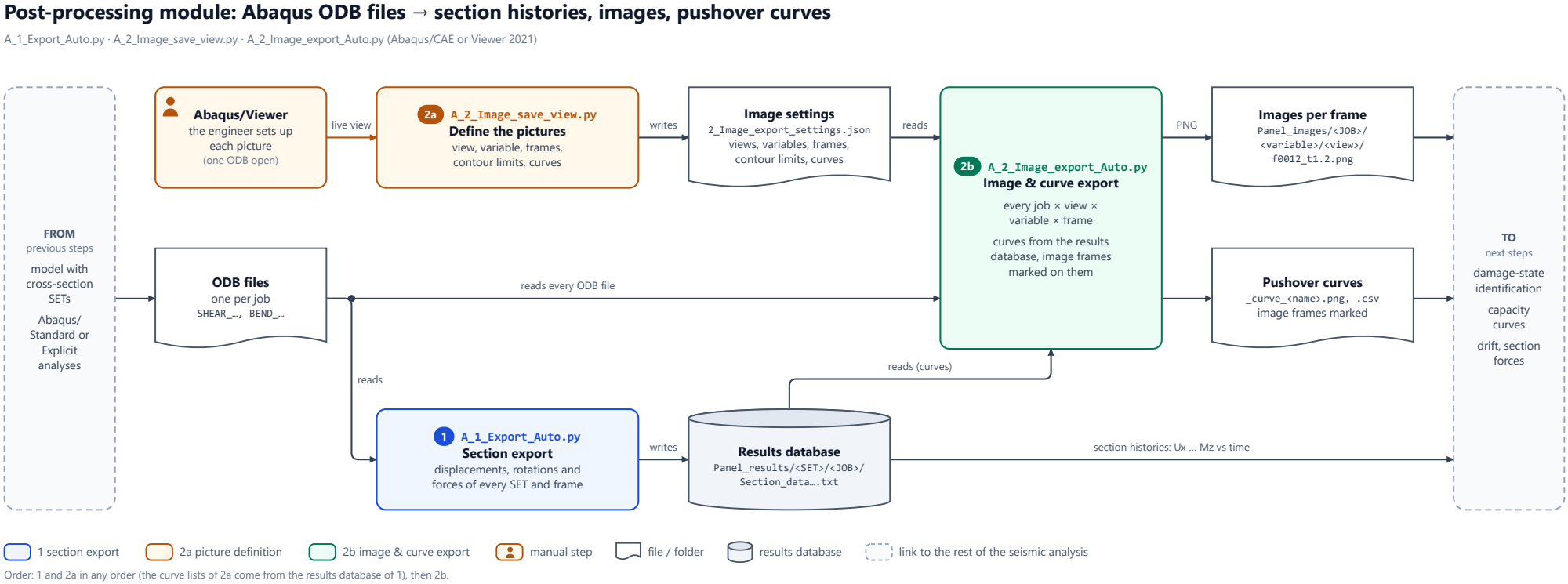


Figure 1 – The module in the seismic analysis. ODB files from the analyses go in; the results database (section histories), images per frame and pushover curves come out. Scripts 1 and 2a can run in either order; 2b uses what both of them stored.

GOES IN

ODB files from the Abaqus analyses (one per job, Standard or Explicit), with the cross-sections defined as node and element SETs named `TYPE&LABEL&INDEX&ANGLE` ; for 2a one ODB opened in Abaqus/Viewer.

HAPPENS

1 displacements, rotations and forces of every section and frame; **2a** pictures defined once, interactively; **2b** every picture printed for every job and frame, curves drawn.

COMES OUT

Section histories (Ux, Uy, Uz, fix, fiy, fiz, Fx, Fy, Fz, Mx, My, Mz vs time) per SET and job; PNG images per job, variable, view and frame; pushover curves (PNG + CSV) with the image frames marked.

Detailed flow

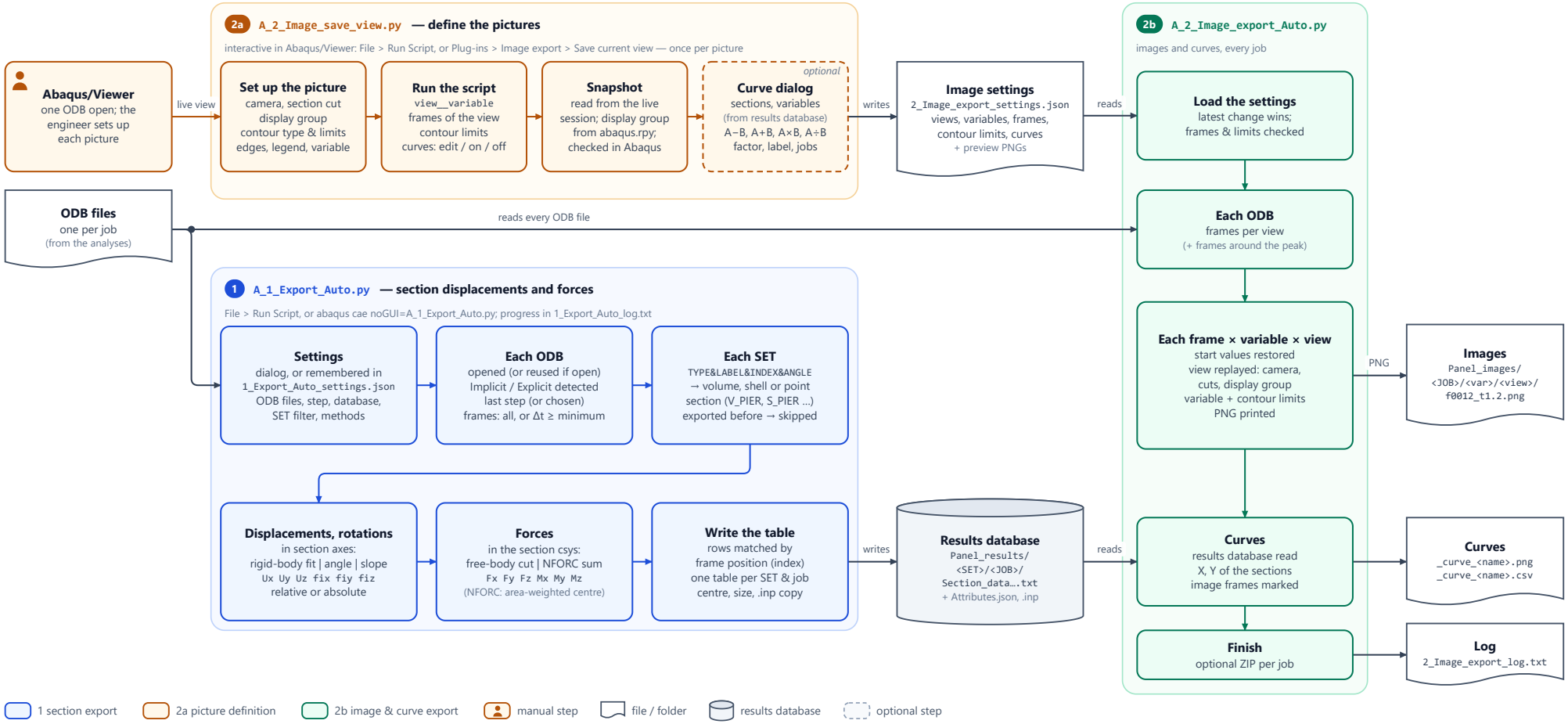


Figure 2 – What each script does. Coloured arrows are the steps inside a script; dark arrows are data read from or written to files. A dashed box is optional.

The three scripts

	Script / how to run	Reads	Does	Writes
1	**`A_1_Export_Auto.py`** File > Run Script (or `abaqus cae noGUI=A_1_Export_Auto.py`); settings dialog first	ODB files of the folder; `1_Export_Auto_settings.json` (last settings)	Per ODB: Implicit/Explicit detected, last step, frames (all, or a minimum time step). Per cross-section SET: section type from the name; displacements and rotations in the section axes (rigid-body fit, angle or slope; relative or absolute); forces in the section csys (Abaqus free-body cut or NFORC summation); rows matched by frame position.	`Panel_results/<SET>/<JOB>/Section_data….txt`, `Attributes.json`, `.inp` copy, job list; settings and log file
2a	**`A_2_Image_save_view.py`** in Abaqus/Viewer with the picture set up: File > Run Script, or Plug-ins > Image export > Save current view	the live Viewer session; `abaqus.rpy` (display group); results database (curve lists)	Snapshot of the picture: camera, section cuts, display group, contour type and limits, edges, legend, variable. Frames and contour limits per view; curve definitions (section variables, A−B, A+B, A×B, A÷B, factor) in a dialog with drop-down lists.	`2_Image_export_settings.json`; previews in `Panel_images/_view_previews`
2b	**`A_2_Image_export_Auto.py`** File > Run Script (MODE = "export")	ODB files; `2_Image_export_settings.json`; results database (curves)	Per ODB: frames per view (evenly spread, all, every n-th, frame numbers, times; optionally frames around the peak). Per frame × variable × view: start values restored, view replayed, variable and contour limits set, PNG printed. Curves of the job with the image frames marked. Optional ZIP per job.	`Panel_images/<JOB>/<variable>/<view>/f0012_t1.2.png`, `_curve_<name>.png / .csv`; settings and log file

Output folders

```
Panel_results/                          results database  (A_1_Export_Auto.py)
  baza sadzaj liste.txt                 SET names and exported jobs
  <SET>/Attributes.json                 section centre and dimensions
  <SET>/<JOB>/Section_dataIIII<SET>IIII<JOB>.txt
        Time, frameId, Ux, Uy, Uz, fix, fiy, fiz, Fx, Fy, Fz, Mx, My, Mz
  <SET>/<JOB>/<job>.inp                 copy of the input file

Panel_images/                           images  (A_2_Image_export_Auto.py)
  <JOB>/<variable>/<view>/f0012_t1.2.png      frame 12, step time 1.2
  <JOB>/_curve_<name>.png  and  .csv          curve, image frames marked
  _view_previews/                           previews  (A_2_Image_save_view.py)
```

Section types (SET names)

TYPE	geometry	axes	forces
`V_PIER, V_SPEC`	volume (solid elements)	pier	yes
`S_PIER`	shell (line of nodes)	pier	yes
`V_SPANDREL`	volume	spandrel	yes
`S_SPANDREL`	shell	spandrel	yes
`S_DRIFT`	point (nodal average)	global axes	no
`SRIFT`	volume	pier	no

Name: `TYPE&LABEL&INDEX&ANGLE`, e.g. `V_PIER&RU&2&0`. ANGLE turns the section axes about global Z; INDEX only numbers the section.

Using the module in another folder

> Copy everything whose name starts with **A_**: `A_1_Export_Auto.py`, `A_2_Image_export_Auto.py`, `A_2_Image_save_view.py` and the folder `A_IMPORT_additional_functions`. Optionally the two `*_settings.json` files, to keep the saved settings (views, limits, curves). Order: 1 → 2a → 2b; 1 and 2b run unattended over all ODB files of the folder.

Generated 10 November 2026 from the scripts in 3_export_and_image_export. Versions of this flowchart: HTML, PDF report, SVG / PNG / PDF per figure, Visio and draw.io (editable).